Ram Representations and Local Certificates for the Dimension-Six MUB Exclusion

Dr. Muralidhara Subbarao
Technical Report Assisted by AI Models
www.integralresearch.net
Stony Brook University
Technical Report TR-Ram-OpenAI-722-M536

October 7, 2026

Confidential technical report. Private corpus passages remain local.

Abstract

We audit the pinned manuscript *The maximum number of mutually unbiased bases in dimension six*, separating its geometric cover, binary64 arithmetic contract, and reported complete execution. We prove local refinements specific to its phase residual and compatibility tests: a coordinate-covariant physical Gram formulation of normal-displacement certificates, an edge-Laplacian bound for the phase Taylor remainder, and joint support inequalities retaining the common normal as well as tangent displacement of the second basis. The latter extend from a pair of vector balls to an entire proposed two-basis configuration. Explicit Taylor remainders and exact Bernstein conversion give an additional whole-panel exclusion certificate, with a rationally checked Fourier-panel example. We prove a closed-form refinement-distance bound, identify the missing compatibility condition in an attempted rectangular root correction, and give a counterexample to the invalid inference. The private Ram material supplies representations, metric accounting, and remainder discipline; the underlying Taylor, Gram, convex-support, and fixed-point techniques are classical. No global exclusion run, historical novelty, independent verification of the source theorem, or measured computational speedup is claimed. Contribution assessment: C3/5, proved shared-displacement and phase-cover refinements provide notable local certificate gains while full execution verification and demonstrated global exclusion gains remain absent.

Contents

1 Pinned Problem and Exact Conventions **3**
1.1 The object being audited . . . 3
1.2 Quantifiers, normalization, and gauge . . . 3
1.3 Attainment and the conditional upper bound . . . 4

2 Source-Wide Dependency Audit **4**
2.1 Seed cover and fixed-frame completion . . . 4
2.2 Phase balls, refinement, and exclusion . . . 5
2.3 Arithmetic and execution obligations . . . 6

3 Actual Ram Correspondences **7**
3.1 Coordinate conversion and its limits . . . 7
3.2 Taylor jets require remainders . . . 7
3.3 Physical Gram and omitted residual . . . 8
3.4 Whole-panel bounds rather than sampled signs . . . 8

4 Physical Gram Form of the Phase Certificate **8**
4.1 An explicit horizontal synthesis . . . 8
4.2 Coordinate-covariant necessary bounds . . . 9

5 An Edge-Laplacian Phase Remainder **10**
5.1 Source-specific nonlinear account . . . 10
5.2 A strengthened universal six-row constant . . . 10
5.3 An exact strict example and its limits . . . 11

6 Shared-Displacement Compatibility Refinements **12**
6.1 Single-vector nonlinear error without the normal split . . . 12
6.2 Horizontal and shared-normal support . . . 12
6.3 Exact comparison with the source pair test . . . 13

7 Validated Taylor and Bernstein Phase Panels **14**
7.1 A finite Ram moment bank on a phase panel . . . 14
7.2 Whole-box conversion theorem . . . 15
7.3 An exact M536 Fourier-panel exclusion . . . 15

8 Cover Refinement and Rectangular Correction **16**
8.1 Sharper chord-to-angle conversion . . . 16
8.2 The exact normal-operator norm on admitted Roots . . . 17
8.3 A closed-form relation bound . . . 17
8.4 Closed covers and fixed alignment . . . 18
8.5 Why the rectangular map does not prove a root . . . 18

9 Verification, Comparison, and Scope **19**
9.1 Bounded checks and what they establish . . . 19
9.2 Rounding contract for the new tests . . . 19
9.3 Public comparisons current to the report date . . . 19
9.4 Realized contribution and future work . . . 20

10 Conclusion **21**

A Read Scope and Provenance **21**

B Check Protocol and Results **21**

1 Pinned Problem and Exact Conventions

1.1 The object being audited

The catalogue identifier is 536, family 266. The source title is *The maximum number of mutually unbiased bases in dimension six*, attributed to OpenAI and dated September 24, 2026. The catalogue pin is

`adc7f1241b42e322a6451854ab7e4b4c146bf78a`.

The local manuscript root, denoted below by S, is the exact path

`C:/Users/mural/OneDrive/Documents/OpenAI-722-math/_sources/openai-math/preprints/`
`The-maximum-number-of-mutually-unbiased-bases-in-dimension-six-September-24-2026`

Source labels in this report refer to files below S/`build`, not to identically numbered results in this report. The source introduction claims $N(6) = 3$ under its arithmetic conditions and its documented October 4 execution. Our object of analysis is that pinned claim and its local interfaces, not an assertion that the supplied extraction authenticates the claimed run.

1.2 Quantifiers, normalization, and gauge

Use $\langle u, v\rangle = \sum_{j=0}^{5} \overline{u_j} v_j$ on $\mathbb{C}^6$. An orthonormal basis is the column set of a unitary matrix. Two such bases are unbiased if every cross squared inner product is $1/6$. To exclude four bases one must exclude, for *every* $H \in \mathbb{C}^{6\times 6}$ satisfying

$$|H_{ij}| = 1, \qquad HH^* = 6I_6, \tag{1.1}$$

the existence of unitary U, V whose entries all have modulus $1/\sqrt{6}$ and for which every entry of $(H/\sqrt{6})^*U$, $(H/\sqrt{6})^*V$, and U^*V has modulus $1/\sqrt{6}$. The matrix H is not restricted to a classified family. The lower bound of three is logically separate.

Put $\mathrm{e}(t) = \exp(2\pi i t)$. A phase lift $P \in \mathbb{R}^{6\times 6}$ represents $H = \mathrm{e}(P)$ entrywise. Phases are in *cycles* except where a radian variable is explicitly introduced. Entrywise integers leave H unchanged. Row and column phases, row and column permutations, conjugation, and transposition preserve extendibility. For transposition the proof acts on an *exact* unitary $H/\sqrt{6}$: apply its adjoint to all bases, interchange the first two, and conjugate. It must not be applied as a unitary operation at an inexact center.

For $C_6 = I_6 - \mathbf{1}\mathbf{1}^T/6$, let

$$\Pi X = C_6 X C_6, \qquad \mathcal{V} = \{X : X\mathbf{1} = 0,\ \mathbf{1}^T X = 0\}, \qquad \dim \mathcal{V} = 25. \tag{1.2}$$

The ambient real phase space is $\mathbb{R}^{36}$ with row-major indexing, real Frobenius inner product, and norm $\|\cdot\|_F$. The map Π is an orthogonal projection; its kernel consists exactly of row plus column phase arrays. It is not the unwrapped dephasing map $D_0(P)_{ij} = P_{ij} - P_{i0} - P_{0j} + P_{00}$, which is not orthogonal. One has $\Pi D_0 P = \Pi P$. All balls and bins in this report are closed. The phase torus has identified faces, not Dirichlet or Neumann boundary data. Every entrywise integer lift used in an alignment is fixed for that alignment; nearest-lift uniqueness is neither assumed nor needed.

Define the source's residual and its real derivative by

$$F_{ij}(P) = \frac{1}{2\pi i} \sum_{g=0}^{5} \mathrm{e}(P_{ig} - P_{jg}), \quad 0 \le j < i \le 5, \tag{1.3}$$

$$(L_P X)_{ij} = \sum_{g=0}^{5} \mathrm{e}(P_{ig} - P_{jg})(X_{ig} - X_{jg}). \tag{1.4}$$

Complex edges are realified in real-then-imaginary order, giving $F : \mathbb{R}^{36} \to \mathbb{R}^{30}$ and $L_P : \mathbb{R}^{36} \to \mathbb{R}^{30}$ with preserved Euclidean norms. Then $F(P) = 0$ if and only if (1.1) holds. The restricted derivative has domain $\mathcal{V}$, of dimension 25, and codomain of dimension 30. No implicit square inverse is available.

1.3 Attainment and the conditional upper bound

For completeness, take the three 2×2 unitary matrices

$$A_0 = I_2, \quad A_1 = 2^{-1/2} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}, \quad A_2 = 2^{-1/2} \begin{pmatrix} 1 & 1 \\ i & -i \end{pmatrix}.$$

With $\omega = \mathrm{e}(1/3)$, set $B_0 = I_3$, $B_1 = (\omega^{jk})_{j,k=0}^{2}/\sqrt{3}$, $B_2 = \mathrm{diag}(1, \omega, \omega)B_1$. The identity $1+\omega+\omega^2 = 0$ proves unitarity. The three-term Gauss sums in $B_1^* B_2$ are $1+2\omega$ or $2+\omega^2$, of squared modulus three; the entries of $A_1^* A_2$ have squared modulus $1/2$. Hence $A_r \otimes B_r$, $r = 0, 1, 2$, are three MUBs. This elementary construction is inherited mathematics, not a report contribution. The source's upper bound also yields its adjoint-closed Cartan consequence, since $\mathrm{tr}((bb^* - I/6)(cc^* - I/6)) = |\langle b, c\rangle|^2 - 1/6$ and the Killing form on $\mathfrak{sl}_6(\mathbb{C})$ is $12\,\mathrm{tr}(XY)$. Without validating the upper-bound premises, that consequence remains conditional as well.

2 Source-Wide Dependency Audit

2.1 Seed cover and fixed-frame completion

The following audit follows all substantive source sections, including the arithmetic appendix. Reading a proof and tracing its dependencies are not the same as independently verifying the entire implementation.

Seed reduction. In `sections/seed.tex`, `seed:arrangement` selects a minimum largest circular gap over row and column differences, uses transposition if needed, and fixes an orientation. Ties and repeated phases are retained. Five phases of the second row and four additional phases of the selected second column give nine coordinates, not ten: the shared entry is counted once. Closed $N = 112$ bins and the sum equations give

$$\|c + J\Delta\| \le 5\pi/(4N^2), \qquad |\Delta_j| \le 1/(2N).$$

Shared-coordinate clipping must use the same midpoint in the two equations. For center reuse, $h = N(q - u) \in [-1/2, 1/2]^9$ and $k = N(u - u')$ obey

$$N^2 \|q - u'\|^2 \le \|k\|^2 + 9/4 + 2k \cdot h.$$

The source's dual support evaluates any finite proposed v using the stacked four-real-equation system. A missed reuse opportunity inserts a midpoint; it is not an exclusion. Radius .032 becomes .0320001 after binary phase conversion. The proposition `seed:cover` needs the *complete* finite traversal, not a list of centers from a partial run.

Stage A completion. In `sections/completion.tex`, use block sizes $2 + 4$:

$$H' = \begin{pmatrix} E' & B' \\ C' & D' \end{pmatrix}.$$

At the seed center, B_0 is 2×4 and C_0 is 4×2. Accepted positive singular lower bounds permit fixed kernel frames $K_B, K_C \in \mathbb{C}^{4\times 2}$ and pseudoinverse factors $G_B = B_0^*(B_0 B_0^*)^{-1}$, $G_C = C_0(C_0^* C_0)^{-1}$. The seed center need not be feasible. The compressed completion $Y = K_C^* D' K_B$ satisfies

$$6I - Y^*Y \succeq 0, \qquad \operatorname{tr}(6I - Y^*Y) \le q = R_s^2 \left(1 + \frac{A_*^2 + 6}{m_C^2}\right), \quad R_s = 2\pi r_s.$$

If $q < 6$, its polar unitary V obeys

$$\left\| Y - \sqrt{6}V \right\|_F \le p = \frac{q}{\sqrt{6} + \sqrt{6 - q}}.$$

This follows by applying the scalar square-root identity to the two positive deficits and bounding their Euclidean norm by their sum. This is a classical polar argument with source-specific defect control, not a Ram discovery.

The closed $U(2)$ parameter domain is $[0, 1/2] \times [0, 1/4] \times [0, 1] \times [0, 1]$ in cycles, for $(\alpha, t, \beta, \gamma)$, with matrix

$$\mathrm{e}(\alpha) \begin{pmatrix} \mathrm{e}(\beta)\cos(2\pi t) & \mathrm{e}(\gamma)\sin(2\pi t) \\ -\mathrm{e}(-\gamma)\sin(2\pi t) & \mathrm{e}(-\beta)\cos(2\pi t) \end{pmatrix}.$$

Zero sine or cosine causes no lost branch, since its phase may be arbitrary. For radian halfwidth $h = \pi/N_u$, the source derives the full-cell bounds $\|V - U\|_F \le \sqrt{6}h$ and $\|V - U - dU[\theta]\|_F \le \sqrt{42}h^2/2$. The comparison block $D = -C_0 E_0^* G_B^* + \sqrt{6} K_C U K_B^*$ has three orthogonal error pieces. Consequently, with $m = \min(m_B, m_C)$,

$$z^2 = \left[R_s(\sqrt{6} + A_*)/m\right]^2 + (6h + p)^2,$$
$$E_r^2 = \left[\frac{R_s^2(L + A_0 + 1)}{2m} + \frac{zR_s}{m}\right]^2 + \left[\frac{\sqrt{252}}{2}h^2 + p\right]^2$$

bound $\|D' - D\|_F^2$ and the squared linearization error respectively. Flatness imposes $\sum(|D_{ij}| - 1)^2 \le z^2$. The directional test stacks four real seed equations and sixteen flatness equations. Its quadratic term uses the *negative part* of the flat weights after moving terms to the right. Replacing it by an unqualified signed maximum would be unsafe. Base grids 16, 32, 64, deeper grids through 512, and the extra-node cap do not prove exhaustion on their own. A live capped cell remains unresolved.

2.2 Phase balls, refinement, and exclusion

Root assertions. In `sections/phase-cover.tex`, a stored $T \in \mathbb{R}^{36\times k}$, $k \le 10$, is only approximately horizontal and orthonormal. For an auxiliary $M \in \mathbb{R}^{36\times 30}$, checked bounds on

$$\Pi - TT^* - ML_P\Pi, \quad T^*T - I, \quad (I - \Pi)T, \quad F(P)$$

yield, for *every horizontal exact-solution displacement* δ,

$$\|(I - TT^*)\delta\| \le n_0 + \iota \|\delta\| + \Gamma \|\delta\|^2, \qquad \|\delta\|^2 \le (1 + o)\|T^*\delta\|^2 + \|(I - TT^*)\delta\|^2. \tag{2.1}$$

The source's "Root" is a certificate of necessary displacement bounds; it neither asserts $F(P) = 0$ nor asserts an existing nearby root.

The chord identity $|D'_j - D_j|^2 = (|D_j| - 1)^2 + |D_j||D'_j - D_j/|D_j||^2$ subtracts the radial flatness defect before estimating angular displacement. Positive magnitude lower bounds and the guard $q_d < 3.8$ justify its arc bound. Candidate phase conversion retains a cycle correction $\|\epsilon\| \le 2 \cdot 10^{-6}$. Raw tangent calculations must also retain the term $\texttt{hor}\sqrt{R_s^2 + \texttt{arc}^2}$ in radians. The source uses two valid ellipsoidal enclosures and sixteen angular corners; the covariance from stored coefficients has a separate coefficient-error charge. Its resolvent calculation certifies the trial solve by its residual. Finite radius bootstrap is an induction on upper bounds, not a contraction or a root-existence theorem. Atlas merges use an actual signed permutation, optional transpose, and fixed integer alignment with the triangle bound $r + d$.

Stage B coverage. In `sections/refinement.tex`, a parent ball supplies tangent coordinates $\eta = T^*\delta$ in the ball of radius $\sqrt{1 + o_P}R$. Closed boxes cover that range, including $k = 0$. The relation at a child compares the same exact solution in two gauges, with a residual term at the inexact child center. Both parent and child Gram defects enter the distance iteration. The searched tangent threshold is sound because its final positive lower endpoint was tested successfully; monotonicity of rounded evaluations is not required. Whole-base-box marking is independently checked by the farthest corner, even when a relation was found in a small subbox.

The blocker induction permits only *already completely covered* parents to be blockers. Valid child spheres produced during a failed parent may be reused, but the failed parent cannot be omitted. Centers and certificate constants are immutable; only radii increase. Emitting a child at the common cap enlarges its solution set, but the entire batch collection, final count, and empty bad-parent list are still necessary. The three source passes use $(\tau, w) = (.07, .013), (.016, .005), (.011, .0038)$.

Stage C exclusion. In `sections/exclusion.tex`, $r \le .03$ and $\left\|\mathrm{e}(P)/\sqrt{6}\right\|_{\mathrm{op}} < 1.00001 =: K$ follow from the small residual. Five vector phases are binned at resolutions $12, 24, 48, 96$, with global phase removed by subtracting the mean. A halfwidth-a bin has exact bounds $X = \sqrt{29/6}a$, $\|x\|_1 \le 5a$, and $\|x\|_\infty \le 3a/2$. Weighted single-vector and joint pair tests must use one common Hadamard displacement. Clustering is certified by a final complete assignment whose centers no longer move. Ball radii below .1 ensure that the twelve vectors of two further bases have distinct labels. Orthogonality edges and unbiasedness edges are necessary possibilities only, not pairwise realizations at a common matrix. The graph search must exclude two disjoint six-cliques with all 36 cross edges. A graph cap is a surviving possibility, never a negative result.

2.3 Arithmetic and execution obligations

The full appendix `sections/rounding.tex` was read, including underflow, range reduction, seed clipping, fixed frames, Root identity, phase conversion, resolvent cancellation, refinement, single-vector and joint supports, and exceptional-value paths. Its assumptions include binary64 nearest rounding, separate basic operations, correctly rounded square root, no reassociation or contraction, gradual underflow, masked exceptions, and specified integer and record representations. Its constants distinguish ideal mathematical quantities from exact values of stored binary data. An inaccurate proposed direction is harmless only after finite-range checks and a separate valid inequality evaluation.

In particular, the appendix proves a degree-18/19 trigonometric approximation after range reduction, with a common $6 \cdot 10^{-14}$ exponential budget including argument errors. It uses an exact Machin-series certificate for π, not an assumed library sine accuracy. The signed resolvent

upper involves possible cancellation and needs its explicit absolute dot surplus and relative quotient surplus; a table of positive-operation margins is not a substitute. The `pow(x,2)` lowering check applies to the actual binary. It must be repeated for a different build. These are important source dependencies, not independently authenticated compiler facts here.

The execution section reports 249878 seed centers, 3083 coarse records, 600 mid records, 22379 fine records in 88 batches, 163 initial C failures from graph caps, and 1127 fallback records, followed by zero final failures. These are *reported source counts*, not computations of this report. The source explicitly distinguishes the October 4 run from an unauthenticated September 7 run. Its finite-execution implication requires all four A shards, all B parents and batches, all C inputs and saved failures, the fallback subsequence, all final shards, exact parsing, exits, provenance, and arithmetic conditions. The eight-byte zero-record files have meaning only within that complete chain.

The local extraction inspected here contains the manuscript files but not `verification/code/`, `verification/computation/`, the seven named C++ sources, the binary records, or the archived logs. Therefore no claimed source-code correspondence in the manuscript is promoted here to an inspected code fact. No source executable was built or run. The missing artifacts are an evidence limitation of this audit, not a proof that the source theorem is false. All new checks below use small independent report-owned scripts.

3 Actual Ram Correspondences

The following exact local paths identify the private passages used. Each correspondence depends on the displayed mathematical content, not on a routing map. We do not assert that the cited corpus passage already solves the MUB problem.

3.1 Coordinate conversion and its limits

`C:/Users/mural/OneDrive/Documents/RamFoundationTRs/`
`TR-rev-1-old-tr250/TR-rev-1-old-tr250.tex`

The subsections "What a Ram representation means in this report" and "A concrete polynomial Ram construction", equation `eq:moments`, and theorem `thm:transfer` describe $m = Ec$, $c = E^{-1}m$ on a fixed finite physical space, with physical mass transformed by congruence. A rectangular measurement bank reconstructs only compatible data in its range. The passage explicitly separates invertible coordinate transfer from a newly discovered theorem and warns that a coordinate change can worsen conditioning. For M536 the finite physical space is horizontal phase displacement, or a specified finite polynomial panel, not a PDE space. Section 4 uses exactly that synthesis/metric distinction; it makes no continuum-frame claim.

3.2 Taylor jets require remainders

`C:/Users/mural/OneDrive/Documents/RamCalBooks/volumes/`
`volume1-rev2/chapters/ch03-local-moments-rev2.tex`

Theorem `thm:ram-local`, equations `eq:ram-integral-remainder` and `eq:ram-remainder-bound`, retain the $(N+1)$st derivative and absolute moment in a localized Taylor representation. The section "Polynomial exactness is not polynomial closure", theorem `thm:degree-closure`, and equation `eq:ram-jet-product` distinguish truncation, closed finite spaces, and differentiated coefficient terms. Here $\mathrm{e}(t)$ has every real derivative of known modulus $(2\pi)^j$. Section 7 derives the finite

exponential-sum remainder directly. We use the corpus's remainder discipline, not an unjustified claim that a trigonometric residual is a finite polynomial or that the original integral-kernel theorem applies without adaptation to a discrete atom bank.

3.3 Physical Gram and omitted residual

`C:/Users/mural/OneDrive/Documents/RamCalBooks/volumes/volume21/chapters/ch01d-physical-metrics.tex`

The entire passage was read. Its exact labels are

`v21c01:physical-gram-definition`, `v21c01:metric-symmetry`, `v21c01:residual-decomposition`, `v21c01:residual-identity`, and `v21c01:gram-equivalence`.

They supply $G = J^*J$, the orthogonal projector $JG^{-1}J^*$, the omitted-residual identity, and relative metric bounds. For M536 a small tangent-coordinate residual cannot erase normal displacement; nor can a zero corrected residual erase residual components outside the correction map. Sections 4 and 8.5 make those statements quantitative in the actual 36/30/25-dimensional spaces.

3.4 Whole-panel bounds rather than sampled signs

`C:/Users/mural/OneDrive/Documents/RamCalBooks/volumes/volume16/chapters/ch05d-clearance.tex`

The section "Polynomial enclosures in Bernstein form", labels `v16c05:powerbernstein`, `v16c05:conversionproof`, `v16c05:hull`, the subsection "Subdivision without introducing approximation", and "A Taylor remainder around a nonpolynomial constraint" give exact basis conversion and the separate interval remainder. The application here is a phase cell rather than a trajectory. A sign certificate must dominate both polynomial enclosure error and trigonometric remainder; depth exhaustion is unresolved. The transfer preserves this exact logical rule.

4 Physical Gram Form of the Phase Certificate

4.1 An explicit horizontal synthesis

Let $a_i = e_i - e_0 \in \mathbb{R}^6$ for $1 \le i \le 5$. The 25 matrices $a_i a_j^T$ form a basis of $\mathcal{V}$. For $\mathcal{J}c = \sum_{i,j=1}^{5} c_{ij} a_i a_j^T$, the physical Gram is

$$G_{\mathcal{J}} = (I_5 + \mathbf{1}\mathbf{1}^T) \otimes (I_5 + \mathbf{1}\mathbf{1}^T). \tag{4.1}$$

Indeed $a_i^T a_j = \delta_{ij} + 1$, so tensor inner products give this formula. Its eigenvalues are 1 (multiplicity 16), 6 (multiplicity 8), and 36 (multiplicity 1). Thus its Euclidean condition number is 36; the physical norm is $\sqrt{c^T G_{\mathcal{J}} c}$, not $\|c\|_2$. Because $\mathcal{J}$ is injective with range $\mathcal{V}$, $\Pi = \mathcal{J} G_{\mathcal{J}}^{-1} \mathcal{J}^*$. This rational identity gives a fully specified finite Ram-compatible coordinate bank. Taking $m = Ec$ for any declared invertible moment matrix transports the Gram to $E^{-T} G_{\mathcal{J}} E^{-1}$. It changes no solution set.

4.2 Coordinate-covariant necessary bounds

Theorem 4.1 (Physical Gram normal certificate)**.** *Let $S : \mathbb{R}^k \to \mathcal{V}$ be injective, $G = S^*S \succ 0$, and $Q = SG^{-1}S^*$. Let $M : \mathbb{R}^{30} \to \mathbb{R}^{36}$ be any fixed real map and suppose*

$$\|(\Pi - Q) - ML_P\Pi\|_{\mathrm{op}} \le \alpha, \quad \|MF(P)\| \le n_0, \quad \|M\|_{\mathrm{op}} \le \mu.$$

Assume, for all $x, v \in \mathcal{V}$, the remainder $\mathcal{R}_P(x) = F(P+x) - F(P) - L_P x$ satisfies

$$\|M\mathcal{R}_P(x)\| \le C\|x\|^2,$$
$$\|M(\mathcal{R}_P(x) - \mathcal{R}_P(v))\| \le C\|x - v\|(\|x\| + \|v\|).$$

Then for every $\delta \in \mathcal{V}$ with $F(P+\delta) = 0$, defining $a = G^{-1}S^*\delta$ and $s = \|\delta\|$, one has

$$\|(I - Q)\delta\| \le n_0 + \alpha s + Cs^2, \tag{4.2}$$
$$s^2 = a^T G a + \|(I - Q)\delta\|^2, \tag{4.3}$$
$$\|(I - Q)(\delta - v)\| \le [\alpha + C(s + \|v\|)]\,\|\delta - v\| + \mu\,\|F(P+v)\| \quad (v \in \mathcal{V}). \tag{4.4}$$

All assertions remain identical after S is replaced by SR for any invertible real $R \in \mathbb{R}^{k\times k}$, with coordinates and Gram transformed accordingly. The case $k = 0$ uses $Q = 0$ and zero tangent energy.

Proof. The matrix Q is the orthogonal projector onto $\operatorname{ran} S$: $Q^* = Q$, $Q^2 = Q$, and $QS = S$. Set $E = (\Pi - Q) - ML_P\Pi$. Since $\delta = \Pi\delta$ and $F(P+\delta) = 0$,

$$(I - Q)\delta = E\delta - MF(P) - M\mathcal{R}_P(\delta).$$

Taking norms gives (4.2). The decomposition $\delta = Sa + (I - Q)\delta$ is orthogonal, proving (4.3). Subtract the Taylor equations at δ and v to obtain

$$(I - Q)(\delta - v) = E(\delta - v) - MF(P+v) - M(\mathcal{R}_P(\delta) - \mathcal{R}_P(v)),$$

which proves (4.4). Under $S \mapsto SR$, $G \mapsto R^T G R$, $a \mapsto R^{-1}a$, and Q and $a^T G a$ are unchanged. □

This theorem sharpens the *representation* of the source's near-orthogonal splitting: it replaces a Euclidean tangent surrogate by exact physical energy. It does not claim that C or α automatically become smaller. A numerical implementation needs certified lower Gram bounds and inverse residuals. For example, if $\left\|G - \widehat{G}\right\| \le \epsilon < \widehat{g}$ and $\widehat{G} \succeq \widehat{g}I$, then $G \succeq gI$ for $g = \widehat{g} - \epsilon > 0$. For any trial coefficient z and $b = S^*\delta$, the exact residual identity gives the interval

$$b^T G^{-1} b = b^T z + b^T G^{-1}(b - Gz), \qquad |b^T G^{-1} b - b^T z| \le \|b\|\,\|b - Gz\|/g. \tag{4.5}$$

The residual and dot must themselves be enclosed outward.

Proposition 4.2 (Conversion from a stored tangent frame)**.** *If the source frame satisfies $\|T^*T - I\| \le o$ and $\|(I - \Pi)T\| \le h$ with $o + h^2 < 1$, then $S = \Pi T$ is injective, $\|S^*S - I\| \le o + h^2$, and its orthogonal projector satisfies*

$$\|Q - TT^*\| \le o + h^2 + 2h\sqrt{1+o}. \tag{4.6}$$

Thus the source identity bound `id` *implies a valid Gram identity bound $\alpha = \mathtt{id} + o + h^2 + 2h\sqrt{1+o}$ using the same M.*

Proof. Write $T = S + V_0$ with $V_0 = (I - \Pi)T$. Orthogonality gives $S^*S = T^*T - V_0^*V_0$, proving the Gram claim and injectivity. The nonzero eigenvalues of SS^* equal those of S^*S, and its range equals that of Q, so $\|Q - SS^*\| = \|I - S^*S\| \le o + h^2$. Also $TT^* - SS^* = V_0T^* + SV_0^*$, of norm at most $2h\sqrt{1+o}$. The triangle inequality proves both conclusions. □

This provides a valid migration path without silently treating stored T as an exact orthogonal matrix. Rechecking the full identity for Q may be sharper than this conservative transfer.

5 An Edge-Laplacian Phase Remainder

5.1 Source-specific nonlinear account

For one scalar cycle displacement define $r(t) = (\mathrm{e}(t) - 1 - 2\pi i t)/(2\pi i)$. Since $r'(t) = \mathrm{e}(t) - 1$, integration and $|\mathrm{e}(t) - 1| \le 2\pi|t|$ give

$$|r(t)| \le \pi t^2, \qquad |r(t) - r(v)| \le \pi|t - v|(|t| + |v|). \tag{5.1}$$

For the second inequality integrate on the segment between t and v and use convexity of absolute value. If $a_{ij}(x) = \|x_i - x_j\|^2$, summation and Cauchy–Schwarz give

$$|\mathcal{R}_{P,ij}(x)| \le \pi a_{ij}(x), \tag{5.2}$$

$$|\mathcal{R}_{P,ij}(x) - \mathcal{R}_{P,ij}(v)| \le \pi\sqrt{a_{ij}(x - v)}\big(\sqrt{a_{ij}(x)} + \sqrt{a_{ij}(v)}\big). \tag{5.3}$$

These hold for all real displacements, not only for small cells.

Theorem 5.1 (Graph-Laplacian remainder certificate)**.** *Split any real $M : \mathbb{R}^{30} \to \mathbb{R}^{36}$ into two-column edge blocks M_e. Choose certified numbers $w_e \ge \|M_e\|_{\mathrm{op}}$, and set*

$$L_w = \sum_{e=(i,j),\, i>j} w_e (e_i - e_j)(e_i - e_j)^T. \tag{5.4}$$

If $b_L \ge \lambda_{\max}(L_w|_{\mathbf{1}^\perp})$, then $C = \pi b_L$ is valid for both remainder bounds in Theorem 4.1 and for the source's `phase:root-bounds`*. Consequently it may be minimized with the source bounds $\pi\sqrt{8}\,\|M\|$ and $\pi\sqrt{6}\,\|\mathcal{S}\|$.*

Proof. For a unit output vector y, the edge norm of M^*y is $\|M_e^* y\| \le w_e$. Since each column $x_{:,g}$ of a horizontal x has zero sum,

$$\sum_e \|M_e^* y\|\, a_e(x) \le \sum_g x_{:,g}^T L_w x_{:,g} \le b_L \|x\|_F^2 .$$

Pairing (5.2) with M^*y proves the first remainder bound. For the difference, apply Cauchy–Schwarz with the nonnegative edge weights $\|M_e^* y\|$ separately to $\sqrt{a_e(x - v)a_e(x)}$ and $\sqrt{a_e(x - v)a_e(v)}$. The resulting upper bound is $\pi b_L \|x - v\|\,(\|x\| + \|v\|)$. Supremum over unit y proves the claim. Each of the three candidate constants proves the same pair of inequalities; their minimum is therefore valid. □

5.2 A strengthened universal six-row constant

The independent bounded mathematical audit supplied the following improvement; we include its full proof so that this report is self-contained. It is a valid refinement of the source inequality, not a historical novelty claim.

Proposition 5.2 (Six-row fourth-power bound)**.** *For every horizontal real 6×6 matrix x,*

$$\sum_{i>j} \|x_i - x_j\|^4 \le \frac{22}{3} \|x\|_F^4 .$$

Consequently $C = \pi\sqrt{22/3}\,\|M\|$ controls both the quadratic and difference remainders after any real map M. This improves the source's first candidate $\pi\sqrt{8}\,\|M\|$ without additional rank assumptions.

Proof. For $s = \|x\|_F > 0$ put $u_i = \|x_i\|^2/s^2$. Since $\sum_i x_i = 0$, Cauchy–Schwarz gives $u_i \le 5(1-u_i)$, hence $m = \max_i u_i \le 5/6$, and $\sum_i u_i = 1$. If $m \le 1/2$, then $\sum_i u_i^2 \le 1/2$. Otherwise isolating a maximum term gives

$$\sum_i u_i^2 \le m^2 + (1-m)^2 \le (5/6)^2 + (1/6)^2 = 13/18,$$

because this quadratic is increasing on $[1/2, 5/6]$. Expanding the distances, using $\sum_i x_i = 0$, gives the exact identity

$$\sum_{i>j} \|x_i - x_j\|^4 = 6\sum_i \|x_i\|^4 + s^4 + 2\sum_{i,j}(x_i \cdot x_j)^2.$$

The last double sum is at most s^4 by Cauchy–Schwarz, proving $6(13/18) + 1 + 2 = 22/3$. The case $s = 0$ is immediate. For unit y, the edge Cauchy–Schwarz bound now gives

$$\sum_e \|M_e^* y\|\, a_e(x) \le \|M^* y\| \left(\sum_e a_e(x)^2\right)^{1/2} \le \sqrt{22/3}\, \|M\|\, \|x\|^2 .$$

The weighted proof of Theorem 5.1 then yields both remainders. □

Thus a valid combined choice is

$$C = \pi \min\{\sqrt{22/3}\,\overline{m}, \sqrt{6}\,\overline{s}, b_L\}, \tag{5.5}$$

where $\overline{m} \ge \|M\|$ and $\overline{s} \ge \|\mathcal{S}\|$ are certified. The relative improvement in the first branch is $\sqrt{11/12}$; we claim neither optimality nor a production speedup. The whole source record remains valid with its original larger constant. A changed numerical expression still needs its own outward evaluation and provenance.

5.3 An exact strict example and its limits

Suppose M has only one nonzero edge block, an isometry from $\mathbb{R}^2$ to $\mathbb{R}^{36}$. Then $\|M\| = 1$, L_w has eigenvalues $2, 0, 0, 0, 0, 0$, and $b_L = 2$. The source's first candidate is $\sqrt{8}$; its tensor candidate is $\sqrt{12}$ because the two nonzero rows of $\mathcal{S}$ are orthogonal of norm $\sqrt{2}$. The new candidate is strictly smaller than either. The edge quadratic estimate is sharp on horizontal matrices with rows $x_i = v/\sqrt{2}$, $x_j = -v/\sqrt{2}$, others zero, where $\mathbf{1}^T v = 0$ and $\|v\| = 1$. A rank-one version of M aligned with the second-order complex remainder at $P = 0$ also approaches the constant 2π as the displacement tends to zero. Thus the strict comparison is not just an eigensolver artifact.

This example is a remainder map, not an assertion that such a map supplies a complete valid Root certificate at an actual Hadamard. For a true Root one must also check the full projector identity and center residual. If a certified Root admits b_L below both existing candidates, replacing Γ by πb_L decreases the normal polynomial for every positive radius and preserves both difference and absolute bounds. This is the exact conditional interface through which smaller local remainders may reduce cover work. No frequency of improvement among the absent production Records is asserted.

For implementation, each $M_e^* M_e$ is 2×2. A rational spectral upper can be verified by positive semidefiniteness. The six-dimensional test $b_L C_6 - L_w \succeq 0$ similarly certifies the Laplacian bound. Gershgorin supplies a simpler, sometimes weaker alternative. These are finite Gram checks; numerical eigenvalues alone are not outward bounds.

6 Shared-Displacement Compatibility Refinements

6.1 Single-vector nonlinear error without the normal split

Fix a source Record of radius $r \le .03$. Let $\delta \in \mathcal{V}$, $\|\delta\| \le r$, and $F(P+\delta) = 0$. For each vector ball t, choose its phase center $u^t \in \mathbb{R}^6$ and radius s_t. Its representatives have phases $u^t + x_t$, with $\mathbf{1}^T x_t = 0$, $\|x_t\| \le s_t$, up to irrelevant global phase. Set

$$\begin{aligned} q^t_{ji} &= \mathrm{e}(u^t_j - P_{ji})/\sqrt{6}, & S^t_i &= \sum_j q^t_{ji}, \\ c^t_i &= (|S^t_i|^2 - 1)/(4\pi), & J^t_{ij} &= -\Im(\overline{S^t_i} q^t_{ji}). \end{aligned} \tag{6.1}$$

For weights $y_t \in \mathbb{R}^6$, define a phase array $(H_t)_{ji} = y_{t,i} J^t_{ij}$ and the nonnegative bound

$$\begin{aligned} E_t(y_t) = \pi \Bigg[&\|y_t\|_\infty (Ks_t + r)^2 + s_t^2 \max_j \left| \sum_i y_{t,i} \overline{S^t_i} q^t_{ji} \right| \\ &+ \frac{2}{\sqrt{6}} \left\| S^t \circ y_t \right\| s_t r + \frac{1}{\sqrt{6}} \left\| S^t \circ y_t \right\|_\infty r^2 \Bigg]. \end{aligned} \tag{6.2}$$

Here K is any proved upper bound on $\left\| \mathrm{e}(P)/\sqrt{6} \right\|_{\mathrm{op}}$, such as the source's 1.00001. If the represented vector is unbiased to this one second basis, then

$$\left| y_t \cdot c^t + (J^{t*} y_t) \cdot x_t - \langle H_t, \delta \rangle_F \right| \le E_t(y_t). \tag{6.3}$$

To verify the entire remainder, write $S'_i = \sum_j q_{ji} \mathrm{e}(x_j - \delta_{ji})$ and $\Delta_i = S'_i - S_i$. First perturb x, then δ; the chord bound gives $\|\Delta\| \le 2\pi(Ks_t + r)$. The quadratic contribution to $(|S'_i|^2 - 1)/(4\pi)$ is therefore the first term of (6.2). The exact factorization

$$\begin{aligned} \mathrm{e}(x-d) - 1 - 2\pi i(x-d) = &[\mathrm{e}(x) - 1 - 2\pi i x] + [\mathrm{e}(-d) - 1 + 2\pi i d] \\ &+ (\mathrm{e}(x) - 1)(\mathrm{e}(-d) - 1) \end{aligned}$$

and $|\mathrm{e}(t) - 1 - 2\pi i t| \le 2\pi^2 t^2$ give respectively the pure-vector, pure-Hadamard, and mixed terms in (6.2), by summation and Cauchy–Schwarz. The linear coefficient is exactly J_{ij} by differentiating the squared modulus. This proves (6.3) without assigning independent normal errors to different vectors.

6.2 Horizontal and shared-normal support

Write the source splitting as $\eta = T^*\delta$, $\nu = A\delta$, $A = I - TT^*$, with certified $\|\eta\| \le r_H$ and $\|\nu\| \le n$. For a real phase array H, define

$$\mathcal{H}(H) = \min \left\{ r \|\Pi H\|, \quad r_H \|T^* H\| + n \|H\|, \quad r_H \|T^* H\| + r \|\Pi A H\| \right\}. \tag{6.4}$$

Each term is an upper for $|\langle H, \delta \rangle|$: the first uses horizontality; the second uses $\delta = T\eta + \nu$; the third additionally uses $\langle H, \nu \rangle = \langle AH, \delta \rangle = \langle \Pi AH, \delta \rangle$. No orthogonality of T or horizontality of ν is presumed. With the exact Gram projector of Theorem 4.1, an additional valid choice is

$$r \|QH\| + n_G \|(\Pi - Q)H\|, \qquad n_G = n_0 + \alpha r + C r^2, \tag{6.5}$$

because $Q \le \Pi$ is an orthogonal projection and both displacement parts are horizontal. More specialized tangent-energy bounds can replace r in the first term when actually certified.

Theorem 6.1 (Whole-configuration support certificate)**.** *Consider any finite set of m vector balls and any finite set of prescribed pair relations. For an oriented pair $e = (t, u)$ let its linearized relation be*

$$q_e + D_e(x_t - x_u) + \varepsilon_e = 0, \qquad \|\varepsilon_e\| \le E_e,$$

where the relation is realified if complex. Choose arbitrary real weights $y_t \in \mathbb{R}^6$ and z_e of the corresponding relation dimension. Define

$$C_* = \sum_t y_t \cdot c^t + \sum_e z_e \cdot q_e,$$
$$g_t = J^{t*} y_t + \sum_{e=(t,u)} D_e^* z_e - \sum_{e=(u,t)} D_e^* z_e, \qquad H_* = \sum_t H_t.$$

Every simultaneous realization of all vectors unbiased to the same second basis, with all the prescribed pair relations, satisfies

$$|C_*| \le \sum_t s_t \|C_6 g_t\| + \mathcal{H}(H_*) + \sum_t E_t(y_t) + \sum_e E_e \|z_e\| . \tag{6.6}$$

Thus a certified strict violation excludes the whole configuration throughout the parent Hadamard ball. It need not exclude any individual pair.

Proof. Sum the weighted equalities behind (6.3) and the weighted edge equations. Their linear part is $C_* + \sum_t g_t \cdot x_t - \langle H_*, \delta \rangle$. Its residual has absolute value at most the last two sums in (6.6). Since $x_t \perp \mathbf{1}$, $|g_t \cdot x_t| \le s_t \|C_6 g_t\|$. The definition (6.4) bounds the one common Hadamard term. Apply the triangle inequality. All quantifiers are uniform over the closed balls, so the proof also covers their boundaries. □

For an orthogonality edge, with $Q_j = \mathrm{e}(u_j^t - u_j^u)$, $S_e = \sum_j Q_j$ and $r_e = s_t + s_u$, use

$$q_e = (\Re S_e, \Im S_e)^T/(2\pi), \quad (D_e)_{:,j} = (-\Im Q_j, \Re Q_j)^T, \quad E_e = \pi r_e^2.$$

For unbiasedness use

$$q_e = (|S_e|^2/6 - 1)/(4\pi), \quad (D_e)_j = -\Im(\overline{S_e} Q_j)/6, \quad E_e = \pi r_e^2 (1 + |S_e|/6).$$

The same expansion as above gives these source constants: the exponential remainder contributes $\pi |S_e| r_e^2/6$, and the squared increment contributes πr_e^2 in the unbiased case. These are unscaled inner-product sums; using a normalized inner product without changing constants would be wrong.

6.3 Exact comparison with the source pair test

For $m = 2$ with one edge, the source's joint test already combines tangent coefficients before taking their norm. It still bounds normal contributions separately by $n(\|H_0\| + \|H_1\|)$ and vector supports by $s_t \|g_t\|$. Our second support candidate uses

$$r_H \|T^*(H_0 + H_1)\| + n \|H_0 + H_1\| \le r_H \|T^*(H_0 + H_1)\| + n(\|H_0\| + \|H_1\|),$$

and $\|C_6 g_t\| \le \|g_t\|$. The nonlinear error terms have not been reduced using linear cancellation. Therefore the exact-arithmetic new right side is never larger than the source pair right side with the same directions and the same pure-vector error option. All old certified exclusions are preserved if rounding is independently accounted for.

The potential strictness occurs within the actual source formulas. Take $P_{ji} = ji/6$, the Fourier phase matrix, and a vector center with $u_0 = 1/4$ and other phases zero. In column $i = 0$, $S_0 = (5+i)/\sqrt{6}$ and $J_{0,0} = -5/6$, $J_{0,j} = 1/6$ for $j > 0$. For $y = (1, 0, \ldots, 0)$ the phase array H is nonzero with $\|H\| = \sqrt{5/6}$. Two equal centers with weights $y, -y$ have $H_* = 0$: the new common normal term is zero, while the separate normal upper is $2n\sqrt{5/6}$ when $n > 0$. This proves strict improvement of a term in a genuine M536 linearization. It is not an excluded MUB configuration: equal centers may already be rejected by simpler tests, and the other remainders remain. We claim neither a newly excluded production cell nor a measured search reduction from this example.

For twelve vertices, prescribe the 30 within-basis orthogonality relations and 36 cross-basis unbiasedness relations. Theorem 6.1 gives a necessary condition for one simultaneous realization of that graph pattern. This adds a configuration-level analytic test after permissive graph search. All weights may be zero except those of a single existing pair, recovering the pair certificate. A finite list of additional weight proposals is safe: failure to find a violating weight retains the pattern. Exhaustive use of this optional test is not needed for soundness, but any unexamined surviving pattern prevents a negative conclusion based solely on such tests.

7 Validated Taylor and Bernstein Phase Panels

7.1 A finite Ram moment bank on a phase panel

Let $s \in [0,1]^p$ and choose an affine lift $P(s) = P_c + \sum_{a=1}^{p}(s_a - 1/2)V_a$ with fixed real phase arrays V_a. The directions may be horizontal, but the following bound does not require it. For edge (i,j) and column g, put

$$d_{ijg}(s) = \sum_a (s_a - 1/2)(V_{a,ig} - V_{a,jg}), \qquad \rho_{ijg} = \frac{1}{2}\sum_a |V_{a,ig} - V_{a,jg}|.$$

For an integer $N \geq 0$, the explicit polynomial

$$p_{ij,N}(s) = \frac{1}{2\pi i}\sum_{g=0}^{5} \mathrm{e}(P_{c,ig} - P_{c,jg}) \sum_{n=0}^{N} \frac{(2\pi i d_{ijg}(s))^n}{n!} \tag{7.1}$$

satisfies the uniform complex-modulus remainder

$$|F_{ij}(P(s)) - p_{ij,N}(s)| \leq \epsilon_{ij,N} := \frac{(2\pi)^N}{(N+1)!}\sum_g \rho_{ijg}^{N+1}. \tag{7.2}$$

Indeed the one-variable integral Taylor remainder of $\mathrm{e}(d)$ is bounded by $(2\pi|d|)^{N+1}/(N+1)!$ along its real segment; division by 2π and summation give the formula. There is no unproved "small remainder".

For a fixed real edge functional $y = (y_{ij})$ with two components per edge, the real polynomial $p_y = \sum \langle y_{ij}, p_{ij,N}\rangle_{\mathbb{R}^2}$ approximates $\langle y, F(P(s))\rangle$ with error at most

$$\epsilon_y = \sum_{i>j} \|y_{ij}\|\, \epsilon_{ij,N}. \tag{7.3}$$

On the finite polynomial space of coordinate degrees at most N, a concrete Ram bank is $m_\alpha = \int_{[0,1]^p} s^\alpha p_y(s)\, ds$ for the same index range. In monomial coordinates a_β, its matrix is $E_{\alpha\beta} = \prod_a (\alpha_a + \beta_a + 1)^{-1}$. It is positive definite: a nonzero real polynomial has positive squared integral on the cube. Hence $m = Ea$ is invertible on this finite space. This establishes an exact moment representation, not a conditioning or speed advantage. Complex polynomials are handled by their two real parts.

7.2 Whole-box conversion theorem

Theorem 7.1 (Taylor–Bernstein exclusion with all errors retained)**.** *Let $p_y(s) = \sum_{\alpha \le n} a_\alpha s^\alpha$ be a real polynomial on $[0,1]^p$ with tensor degree vector n. Its Bernstein coefficients are*

$$b_\beta = \sum_{\alpha \le \beta} a_\alpha \prod_{a=1}^{p} \frac{\binom{\beta_a}{\alpha_a}}{\binom{n_a}{\alpha_a}}. \tag{7.4}$$

Suppose certified coefficient intervals give $b_\beta \ge \underline{b}_\beta$, and (7.3) holds. If

$$\min_\beta \underline{b}_\beta > \epsilon_y, \tag{7.5}$$

there is no $s \in [0,1]^p$ with $F(P(s)) = 0$. The analogous negative upper certificate is valid. Interval coefficient errors, including errors in π and center exponentials, are additional to the Taylor remainder.

Proof. In one variable the binomial identity gives $s^a = \sum_{b=a}^{n} \binom{b}{a}\binom{n}{a}^{-1} B_b^n(s)$. Taking tensor products proves (7.4). The tensor Bernstein weights are nonnegative and sum to one; hence $p_y(s) \ge \min_\beta \underline{b}_\beta$ on the entire closed cube. The true functional is at least this lower bound minus ϵ_y and is strictly positive under (7.5). A zero of F would make the functional zero, a contradiction. □

Rational polynomial coefficients give rational Bernstein coefficients; the coefficients in (7.1) are generally not rational. One must enclose them, or choose a special algebraic/rationally bounded case. Positive quadrature alone does not provide an injective moment map on an arbitrarily large polynomial space. Likewise, exact conversion does not remove the remainder of a nonpolynomial phase residual.

For an exact rational midpoint subdivision, de Casteljau uses convex averages and introduces no polynomial approximation. A child's true residual still needs a remainder valid on that child. Retaining the parent remainder is safe but may prevent a strict certificate; recentering the Taylor expansion can reduce it. With fixed Taylor order and strictly positive true margin on a compact panel, sufficiently small cells and certified coefficient enclosures of sufficiently small error eventually yield a positive certificate. With zero margin, no such termination claim follows.

7.3 An exact M536 Fourier-panel exclusion

Let $P^F_{ig} = ig/6$, and vary only entry $(1,1)$: $P(t) = P^F + tE_{11}$, where $t \in [1/64, 1/32]$ in cycles. The six Fourier phases in row difference $(1,0)$ initially sum to zero, so

$$F_{10}(P(t)) = \mathrm{e}(1/6)\frac{\mathrm{e}(t)-1}{2\pi i}, \qquad h(t) := \Re\big(\mathrm{e}(-1/6)F_{10}(P(t))\big) = \frac{\sin(2\pi t)}{2\pi}.$$

This is a legitimate one-dimensional M536 phase panel. It is not necessary to project it for the exclusion, because (1.3) is defined on every lift.

For $z \ge 0$, $\sin z \ge z - z^3/6$: integrating $\cos z \ge 1 - z^2/2$, which follows by integrating $|\sin z| \le z$, proves this on the required interval. With $\pi < 22/7$,

$$h(t) \ge p(t) := t - \frac{968}{147}t^3. \tag{7.6}$$

This one-sided polynomial is already a validated remainder enclosure; it is not a claim that $h = p$. Put $t = (1+s)/64$ and $c = 121/4816896$. Then

$$p((1+s)/64) = \frac{1+s}{64} - c(1 + 3s + 3s^2 + s^3).$$

Its degree-three Bernstein coefficients are exactly

$$b_0 = 1/64 - c, \quad b_1 = 1/48 - 2c, \quad b_2 = 5/192 - 4c, \quad b_3 = 1/32 - 8c. \tag{7.7}$$

They all exceed $3/200 = 0.015$ by integer arithmetic. Thus $h(t) > 0.015$ on the entire closed panel, excluding every Hadamard there. The Fourier point $t = 0$, outside this panel, remains a true solution. The example is deliberately bounded and transparent: it verifies the polynomial/remainder/cover interface on actual source equations, not a new obstruction to four MUBs. The report's exact-rational check evaluates these four inequalities and the moment-to-Bernstein conversion.

8 Cover Refinement and Rectangular Correction

8.1 Sharper chord-to-angle conversion

The source's `phase:arc-bounds` uses a maximum derivative of arcsine. Keeping its integral ratio gives a smaller bound with the same hypotheses.

Proposition 8.1 (Sharp aggregate arc upper)**.** *Suppose $m_j \geq m_* > 0$, D'_j has modulus one, $D_j = m_j D^\circ_j$ with $|D^\circ_j| = 1$, and*

$$\sum_j m_j |D'_j - D^\circ_j|^2 \leq \sigma, \qquad 0 \leq q < 4, \qquad \sigma/m_* \leq q.$$

Let $d_j \in [-\pi, \pi]$ be the shortest radian angle between D°_j and D'_j. Set

$$\kappa(q) = \frac{2\arcsin(\sqrt{q}/2)}{\sqrt{q}} \quad (q > 0), \qquad \kappa(0) = 1.$$

Then

$$\|d\| \leq 2\arcsin(\sqrt{q}/2), \qquad \left(\sum_j m_j d_j^2\right)^{1/2} \leq \kappa(q)\sqrt{\sigma}. \tag{8.1}$$

For $q > 0$, $\kappa(q) < (1 - q/4)^{-1/2}$, the source ratio. The first upper is sharp given only the unweighted chord-square bound q.

Proof. Put $c_j = |D'_j - D^\circ_j|$. Then $\sum c_j^2 \leq q < 4$, so every chord is strictly below two and $|d_j| = 2\arcsin(c_j/2)$. The ratio $\arcsin t/t = \int_0^1 (1 - t^2u^2)^{-1/2}\, du$ is increasing for $0 \leq t < 1$. It is strictly less than its endpoint integrand for $t > 0$. Therefore $|d_j| \leq \kappa(q)c_j$, giving both summed bounds and the strict comparison. For sharpness concentrate chord square q in one coordinate and put every other chord equal to zero. If $q = 0$, all chords and angles vanish and the continuous definition suffices. □

At $q = 3$, the new arc upper is $2\pi/3 < 44/21$, whereas the source upper is $\sqrt{12}$. This changes a necessary conversion between complex completion error and real phase radius, not merely a cosmetic coefficient. Its use in the source still requires a verified upper for arcsine or an equivalent certified sine comparison. An ordinary library call is not licensed by the source's arithmetic appendix. All center and horizontal projection errors remain in the subsequent phase bounds.

8.2 The exact normal-operator norm on admitted Roots

The independent audit also supplied this useful strengthening of `refine:approximate-projection`. If $k \le 10 < 36$ and $\|T^*T - I\| \le o < 1$, all nonzero eigenvalues of TT^* lie in $[1-o, 1+o]$, and its remaining eigenvalues are zero. Thus

$$\|I - TT^*\|_{\mathrm{op}} = 1. \tag{8.2}$$

This proof uses symmetry and the genuine Gram certificate, not just field range checks. It does not assert that TT^* or $I - TT^*$ is idempotent. For $k = 0$ the conclusion is immediate as well. Consequently the source normal alternative $(1 + o_P)d$ can be replaced by d. Its projected child factor has the extra upper

$$\|(\Pi S)^*(I - TT^*)\| \le \|\Pi S\| \le \sqrt{1 + o_Q}.$$

Taking its minimum with the independently certified cross-Gram upper preserves the finite distance iteration. The projected child factor must still use ΠS, not S. Small horizontality error alone cannot justify that replacement. The normal contraction bound also gives $r\,\|\Pi A H\| \le r\,\|H\|$ in (6.4), but it does not make the normal displacement disappear.

8.3 A closed-form relation bound

The source's parent–child relation yields, with $d = \|\delta - d_q\|$ and $z = \sqrt{1 + o_P}\,t$,

$$d^2 \le z^2 + \|\nu\|^2, \qquad \|\nu\| \le \lambda d + e,$$

where $\lambda \ge \alpha_P + C_P(R + \|d_q\|)$ and $e \ge \mu_P\,\|F(Q)\|$. Its general finite iteration works even when $\lambda \ge 1$.

Proposition 8.2 (Direct refinement distance certificate)**.** *If $0 \le \lambda < 1$, every such d satisfies*

$$d \le D_{\mathrm{quad}}(z) = \frac{\lambda e + \sqrt{e^2 + (1-\lambda^2)z^2}}{1 - \lambda^2}. \tag{8.3}$$

For a target radius $a \ge 0$, the sufficient whole-box test

$$z^2 + (\lambda a + e)^2 \le a^2 \tag{8.4}$$

implies $d \le a$. These tests may be minimized with independently valid source upper bounds. They do not assert existence of any solution.

Proof. Substitution gives $(1-\lambda^2)d^2 - 2\lambda e d - (e^2 + z^2) \le 0$. The leading coefficient is positive. Its nonnegative root is (8.3), after simplifying the discriminant; the other root is nonpositive. Thus every admissible nonnegative d lies below the displayed root. If (8.4) holds, the same quadratic is nonnegative at a. Unless $e = z = 0$, its only nonnegative zero is the upper root and it is negative below it; hence a is at least that root. When $e = z = 0$, the original inequality already forces $d = 0$. □

The sufficient target test avoids a nearly singular division when useful. It does not apply with $\lambda \ge 1$. For rational $\lambda = 1/2$, $e = 1/10000$, $a = 1/100$, and $z = 1/125$, its left side is $9001/10^8$, strictly below $a^2 = 10000/10^8$. These are relation-certificate parameters, not measured source Records. The result provides a direct acceptance route for a closed tangent box; finite iterative bounds remain useful outside its regime.

8.4 Closed covers and fixed alignment

For centers P, Q, a signed permutation/transpose Φ, and a fixed integer array N, put $d = \Pi(D_0(\Phi P) - D_0 Q - N)$. For every horizontal δ, $Q + d + \Phi\delta$ represents a matrix equivalent to $P + \delta$. Thus $\|\delta\| \le r$ implies the image lies in the closed ball about Q of radius $r + \|d\|$. The transformed solution need not use the same equivalence as another solution assigned elsewhere. An upper computed distance $\widehat{d} \ge \|d\|$ and an outward sum $\widehat{r} \ge r + \widehat{d}$ suffice. A bucket lookup or wrap proposal cannot replace this final metric test.

For a tangent box $\prod_a [u_a - h_a, u_a + h_a]$ and center coordinate η_q, the squared farthest distance is exactly $\sum_a (|u_a - \eta_{q,a}| + h_a)^2$, because the maximizing endpoints may be chosen independently. This is the correct argument for whole-box assignment. With a nonorthogonal Ram coordinate system and Gram G, Euclidean boxes are not physical balls. The function $(\eta - \eta_q)^T G(\eta - \eta_q)$ is convex in each coordinate; its maximum over a closed box occurs at a corner, so all 2^k corners give an exact finite physical-metric upper. Bounding it by $\lambda_{\max}(G)$ times the Euclidean farthest distance is cheaper but possibly looser. Dropping G entirely is not a valid refinement.

8.5 Why the rectangular map does not prove a root

The source map $M : \mathbb{R}^{30} \to \mathbb{R}^{36}$ is used in a residual identity, not as a two-sided inverse. A tempting but invalid step is to show that $x \mapsto x - MF(P + x)$ is a contraction on a horizontal ball and then infer $F(P + x) = 0$ at its fixed point. A fixed point only gives $MF(P + x) = 0$.

Proposition 8.3 (Counterexample in the actual dimensions)**.** *There are affine maps $\mathcal{F} : \mathbb{R}^{25} \to \mathbb{R}^{30}$ and $M : \mathbb{R}^{30} \to \mathbb{R}^{36}$ with $M\mathcal{F}'(x)$ equal to the horizontal synthesis, for which the corrected iteration is a contraction onto one point but $\mathcal{F}$ has no zero.*

Proof. Choose any orthonormal synthesis $U : \mathbb{R}^{25} \to \mathcal{V}$. For $\epsilon > 0$ set $\mathcal{F}(x) = (x_1, \dots, x_{25}, \epsilon, 0, 0, 0, 0)$ and $M(y) = U(y_1, \dots, y_{25})$. In horizontal coordinates the iteration is $x - U^* M\mathcal{F}(x) = 0$ for every x, with contraction constant zero. Its unique fixed point is 0, yet $\|\mathcal{F}(0)\| = \epsilon > 0$. No zero exists. Also $M\mathcal{F}' = U$, exactly, so no numerical conditioning defect accounts for the failure. □

This is a logical counterexample to a proposed generic inference, not an assertion that this affine map is the Hadamard residual. The source does not make that invalid inference: it explicitly disclaims root existence.

Theorem 8.4 (Sufficient corrected root theorem with compatibility)**.** *Let $U : \mathbb{R}^{25} \to \mathcal{V}$ be orthonormal, let $\mathcal{F}(x) = F(P + Ux)$, and let $B \subset \mathbb{R}^{25}$ be a nonempty closed ball of radius ρ about x_0. Fix $A : \mathbb{R}^{30} \to \mathbb{R}^{25}$ and $K(x) = x - A\mathcal{F}(x)$. Suppose on B:*

1. *K is Lipschitz with constant $q < 1$;*
2. *$\|A\mathcal{F}(x_0)\| \le (1 - q)\rho$;*
3. *for every $x \in B$, $A\mathcal{F}(x) = 0$ implies $\mathcal{F}(x) = 0$.*

Then there is exactly one zero of $\mathcal{F}$ in B. A sufficient condition for item 3 is that all $\mathcal{F}(x)$ lie in a fixed subspace $W \subset \mathbb{R}^{30}$ and $A|_W$ is injective. More generally it suffices to prove $\|\mathcal{F}(x)\| \le \kappa \|A\mathcal{F}(x)\|$ on B.

Proof. For $x \in B$, $\|K(x) - x_0\| \le q\|x - x_0\| + \|A\mathcal{F}(x_0)\| \le \rho$. Hence K maps the complete closed ball into itself. The contraction theorem gives a unique fixed point x_*. It satisfies $A\mathcal{F}(x_*) = 0$; item 3 gives $\mathcal{F}(x_*) = 0$. Conversely any zero is a fixed point, so uniqueness follows. Both proposed sufficient conditions imply item 3 directly. □

Neither the source's full-space defect bound nor a small least-squares residual establishes this compatibility condition. Moreover, continuous families of Hadamards can defeat a full 25-coordinate uniqueness argument. One may instead fix tangent parameters and apply a square theorem in an appropriate normal slice, but the residual-dependence identities and slice hypotheses still need proof. No such universal slice is asserted here.

9 Verification, Comparison, and Scope

9.1 Bounded checks and what they establish

The report-owned program `M536-author-checks.py` uses exact rational arithmetic for the physical Gram identity, moment conversion, Bernstein coefficients, closed-distance example, seed-bin vertex bounds, and the rectangular counterexample. It also evaluates fixed-size floating diagnostics of the derivative, quadratic remainder, difference remainder, and common-normal cancellation. The floating diagnostics are regression checks, not interval proofs and not a substitute for the analytic arguments above. No random sample is interpreted as a uniform certificate.

The finite scopes are explicit: 32 vertices for the five-variable centered vector bin, a 36-by-25 rational synthesis, one 6-by-6 edge Laplacian, four Bernstein coefficients on the Fourier panel, and a fixed list of phase perturbation examples. There is no exhaustive 112-bin seed traversal, no $U(2)$ completion atlas, no production Root or Record deserialization, and no source graph exclusion run. The script, output, and source/corpus input hashes are retained in the author audit. Executed results are summarized in Appendix B; exact inequalities in the report have their proofs independently of those numerical diagnostics.

9.2 Rounding contract for the new tests

The source appendix's numerical margins apply to the source expressions, not automatically to (6.4), (6.6), or a new Gram inverse. For any proposed certificate, a simple sufficient discipline is to construct an interval $[C_-, C_+]$ containing its exact constant and a nonnegative upper R_+ for its entire right side. Reject only if $\operatorname{dist}(0, [C_-, C_+]) > R_+$. Equality retains the case. For a signed non-absolute functional, require its lower bound to exceed the certified error or threshold. If several right-side bounds are minimized, each must first be a valid upper for the same exact expression.

For coefficient arrays, norm perturbations can be charged by $|\,\|a\| - \|\widehat{a}\|\,| \le \|a - \widehat{a}\|$, with Frobenius upper bounds for matrix errors. Shared cancellation must be assessed after including the errors of *all* contributing arrays. If $H_* = \sum H_t$ is approximately zero, setting it to zero would be unsound; its actual norm upper includes the sum of input error bounds. The exact Fourier cancellation example has no such uncharged error because it is symbolic.

Validated Cholesky, rational positive-semidefinite tests, or interval spectral methods may certify the Gram and Laplacian bounds. Direct residual checks are preferable to trusting eigensolver accuracy. Failed rank, symmetry, finite-range, enclosure, or margin checks leave the cell unresolved. These rules establish a mathematical implementation contract, not a completed binary64 implementation of every proposed refinement.

9.3 Public comparisons current to the report date

Public searches used only public mathematical titles/topics, not private passages. Primary-source pages were consulted on October 7, 2026. Jaming, Matolcsi, Móra, Szőllősi and Weiner proved that a standard basis together with any member of the two-parameter Fourier family cannot extend to a

quartet [2]; the later discretization program [3] describes the unrestricted finite-search approach. The pinned manuscript extends the claimed scope to arbitrary second bases, using fixed-frame completions and a recorded execution. Our results refine local inequalities within that framework, not its complete enumeration.

The 2026 primary paper on MUB triplets [4] characterizes triplets by Hadamard cubes and retains a conjectured identity sufficient for the dimension-six conclusion. The August 2026 classification preprint [5] claims an exhaustive classification of order-six Hadamards. Classification of individual Hadamards is not itself an exclusion of simultaneous extensions by two bases. Neither public development supplies the absent production artifacts or validates this report's proposed algorithm on every class. No historical novelty priority is inferred from this limited literature comparison.

Method	Concrete advantage here	Limitation or cost
Source near-orthogonal Root	Small stored frame and explicit scalar defects; no manifold assumption	Approximate projector and tangent norm must keep Gram and horizontal errors.
Physical Gram form	Exact Pythagoras and invariance under an invertible Ram coordinate change	Needs a lower Gram bound and certified inverse; not always a smaller final radius.
Edge Laplacian	A six-dimensional, source-specific third remainder candidate with a proved strict example	May be worse for dense maps; production benefit not measured.
Shared-displacement support	Preserves common normal cancellation and zero-mean vector geometry; extends to entire graph patterns	Larger direction search; pure nonlinear remainders remain; source margins must be redone.
Taylor–Bernstein panel	Covers every point with explicit remainder and exact algebraic conversion	Tensor coefficient count can grow as $(N+1)^p$; near-zero signs remain unresolved.
Closed-form relation	A finite analytic radius or target test when $\lambda < 1$	Not applicable to all source relation parameters; iterative bounds may be stronger.

9.4 Realized contribution and future work

The realized gains are the proved domination of the source pair support with shared normal and horizontal projections, its extension to simultaneous configuration weights, a new optional Laplacian remainder certificate with an exact strict comparison, a coordinate-covariant Root reformulation, and an explicit whole-panel Fourier exclusion with a rigorous one-sided remainder. The integrated six-row and normal-operator improvements came from the independent audit and are proved here. The sharper aggregate arc bound, closed-distance formula, and rectangular counterexample clarify two separate cover/correction interfaces. These are notable local contributions relative to the pinned manuscript, supporting C3 rather than a score based on the catalogue's L5 applicability.

They do not establish a new global MUB theorem, authenticate the source execution, prove that a surviving graph pattern is realizable, or demonstrate a runtime gain. The local constructions mostly adapt classical mathematics to the exact phase residual, with private Ram representations making the metric and remainder bookkeeping explicit. C4 is not assigned: no large production family is newly excluded or substantial complete computational bottleneck demonstrably removed

by the evidence available here.

Useful future work is sharply delimited. First, obtain and authenticate the source code and exact run artifacts against their declared hashes before attempting any global verification. Second, implement the new support tests with their own outward arithmetic audit and test a predeclared bounded set of genuine Records, retaining failures. Third, compare pair and whole-configuration certificates at identical directions and errors, separating per-test cost from exclusion benefit. Fourth, test Laplacian and physical-Gram variants on the actual correction maps and retain all coordinate condition estimates. Finally, use parameter-adaptive Taylor panels only with their remainders and cover maps; a finite jet inverse without compatibility is not an alternative to these obligations.

10 Conclusion

The mathematically useful Ram contribution here is not to rename the MUB problem. It is to retain the physical metric, finite-dimensional compatibility, common displacement, and nonlinear remainder in the exact places where a local numerical proposal becomes a certificate. We have proved refinements at those interfaces and exhibited bounded exact checks on actual phase equations. The global dimension-six exclusion remains a source claim whose complete code, arithmetic, and execution premises have not been independently verified in this report.

A Read Scope and Provenance

The substantive source read scope is exhaustive for the supplied manuscript extraction: `paper.tex`; `sections/introduction.tex` (246 lines), `seed.tex` (490), `completion.tex` (715), `phase-cover.tex` (1008), `refinement.tex` (656), `exclusion.tex` (715), `execution.tex` (400), `conclusion.tex` (11), and the mathematical appendix `rounding.tex` (1566). All these section filenames are under `S/build/sections` except `paper.tex`, which is at the build root. Both figures, `references.bib`, and the top-level README were also read. The referenced implementation and run archives were absent from this local directory. All 14 local source files match both the local SHA-256 manifest and their Git blob identifiers in the cached, nontruncated preprints tree. That tree also contains no M536 verification directory. These are checked local metadata consistencies, not remote authentication or verification of the reported execution. File hashes record the exact bytes read; a catalogue pin alone is not asserted to verify a local checkout.

Private read scope was: TR-rev-1-old-tr250, lines 121–268, 455–478, 1125–1136; volume 1 revised chapter 3, lines 1–115 and 194–285; volume 21 physical-metrics passage in full; volume 16 clearance passage, lines 62–141 and 199–248. Exact paths and section labels are in Section 3. We did not claim to read the whole private books. The catalogue row and contribution policy were read directly. Local hashes and absence checks are in `_audit/M536-author-inputs.json`, and the executed bounded checks are in `_audit/M536-author-check-results.json`. The independent local input `_audit/M536-independent-audit.md` supplied the two improvements proved in Proposition 5.2 and (8.2); its evidence-limited execution conclusion agrees with this report. We do not rebrand its bounded checks as an independent execution of the source pipeline. All audit paths in this appendix are relative to `C:/Users/mural/OneDrive/Documents/OpenAI-722-math`.

B Check Protocol and Results

The exact checks are designed to establish the following finite statements: the horizontal synthesis has the claimed Gram and projection; all 32 centered bin vertices obey the source's three norm constants;

the single-edge Laplacian has largest eigenvalue two; the degree-three moment reconstruction and Bernstein conversion reproduce (7.7); all four coefficients exceed 3/200; the rational target-distance example satisfies (8.4); and the rectangular counterexample has exact corrected residual zero but nonzero full residual. The checks have fixed sizes and no unbounded search loop.

The floating diagnostics use fixed phase matrices and fixed displacements to check the derivative normalization, the quadratic and difference bounds, the stated shared-normal cancellation, and the three-basis construction. Their tolerances account only for diagnostic binary floating error; they are not certified outward bounds. The human-readable analytic proofs are the uniform evidence. The final audit records the actual interpreter, script hash, exit status, results, and finite case counts.

The executed bundled-Python run on October 7, 2026 at 23:47:04 UTC exited zero: all 23 checks passed, comprising 15 exact-rational checks, five floating diagnostics, and three provenance checks. The derivative diagnostic used 20 fixed cases and had maximum absolute discrepancy $1.656 \cdot 10^{-9}$. The maximum sampled ratios to the six-row quadratic and difference upper bounds were respectively 0.328 and 0.167 after upward display rounding. The three-basis diagnostic discrepancy was below $2.260 \cdot 10^{-16}$. These decimal diagnostics carry no interval-certification claim. The exact minimum Fourier-panel Bernstein coefficient was 75143/4816896, exceeding 3/200 by 72239/120422400. The six-row identity was additionally checked on 24 deterministic rational horizontal matrices, alongside its universal proof in Proposition 5.2.

References

[1] OpenAI, *The maximum number of mutually unbiased bases in dimension six*, September 24, 2026, catalogue id 536, pinned manuscript and local source path specified in Section 1.

[2] P. Jaming, M. Matolcsi, P. Móra, F. Szőllősi, and M. Weiner, *A generalized Pauli problem and an infinite family of MUB-triplets in dimension 6*, J. Phys. A 42 (2009), 245305. https://arxiv.org/abs/0902.0882v2.

[3] P. Jaming, M. Matolcsi, and P. Móra, *The problem of mutually unbiased bases in dimension 6*, Cryptography and Communications 2 (2010), 211–220. https://arxiv.org/abs/1201.0640v1.

[4] M. Matolcsi, A. K. Matszangosz, D. Varga, and M. Weiner, *Triplets of mutually unbiased bases*, Journal of Algebraic Combinatorics 63 (2026), article 26. https://doi.org/10.1007/s10801-026-01506-x.

[5] M. Cárdenes Wuttig and J. Tindall, *A Complete Classification of Complex Hadamard Matrices of Order Six*, 2026 preprint. https://arxiv.org/abs/2608.18053v2.

[6] Private Ram corpus, four exact paths and substantive section/label locators in Section 3; local access only.